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INTERLOCK FASTENER

Abstract

A fastener device has a proximal head, a shaft extending from the proximal head and having external threads on at least a portion of the shaft, and a chamber extending internally from the proximal head into the shaft. At least a portion of the chamber has internal threads. The external threads have a first handedness, and the internal threads have a second handedness different from the first handedness.

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Background/Summary

CROSS REFERENCE TO OTHER APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/557,170 entitled INTERLOCK SCREW filed Feb. 23, 2024, which is incorporated herein by reference for all purposes.

BACKGROUND OF THE INVENTION

[0002] Surgical screws are the most implanted medical device in the world, with billions of units being implanted in the global population every year. Many of those screws will be removed within 24 months of entering the body. A survey of national inpatient data samples found that 10% of spinal, hip, pelvis, and femur fixations will be removed or revised within 12 months of implantation, with rates increasing up to 80% over subsequent years in certain cases. Surgical screws are frequently damaged during insertion, patient recovery, or while attempting to be removed. With current surgical screws, it is reported that on average, 48% of surgical screws are stripped during insertion with some surgeons stripping up to 90% of the screws they implant. Additionally, many more are stripped during removal or revision surgeries. Removing stripped or otherwise damaged screws during or post-surgery can be a challenging task for surgeons; current methods of damaged screw removal can contribute to the risk of additional complications including, but not limited to, infection, secondary fractures, and metal debris in the operating field. For revision surgeries, current methods of screw removal may require relocation of the entire fixation device, further increasing procedural time and rates of other complications. Additionally, due to nonmagnetic requirements for MRI compatibility, screws are often unreliably secured to a driver via press fit or other insecure techniques. This can result in the fumbling of screws during installation and result in the risk of dropping a screw in an operating room or into the open operating field of a patient, leading to further complications and extended procedural duration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Various embodiments of the invention are disclosed in the following detailed description and the accompanying drawings.

[0004] FIGS. 1A-1C are diagrams illustrating various views of an embodiment of a fastener.

[0005] FIGS. 1D-1E are diagrams illustrating section views of an embodiment of the fastener.

[0006] FIG. 1F is a diagram illustrating example dimensions of an embodiment of the fastener.

[0007] FIG. 2A is a diagram illustrating an embodiment of an interlock driver tool.

[0008] FIG. 2B is a diagram illustrating example dimensions of an interlock driver tool.

[0009] FIGS. 3A-3C are diagrams illustrating examples of engaging an interlock driver tool with a fastener.

[0010] FIG. 4 is a diagram illustrating an embodiment of a head driver tool.

[0011] FIGS. 5A-5C are diagrams illustrating examples of engaging a head driver tool with a fastener.

[0012] FIGS. 6A-6C are diagrams illustrating various views of alternate embodiments of the fastener shown in FIGS. 1A-1F.

[0013] FIG. 7 is a diagram illustrating an embodiment of a head driver tool.

[0014] FIG. 8 is a diagram illustrating an embodiment of an interlock driver tool.

[0015] FIG. 9 is a diagram illustrating an embodiment of a head driver tool.

[0016] FIGS. 10A-10P are diagrams illustrating various views of different embodiments of fasteners with secondary drive features.

[0017] FIG. 11A is a flowchart illustrating an embodiment of a process for inserting a fastener into a subject.

[0018] FIG. 11B is a diagram illustrating an example implementation of at least a portion of the

process of FIG. 11A.

[0019] FIG. 12A is a flowchart illustrating an embodiment of a process for removing a fastener from a subject.

[0020] FIG. 12B is a diagram illustrating an example implementation of at least a portion of the process of FIG. 12A.

DETAILED DESCRIPTION

[0021] The invention can be implemented in numerous ways, including as a process; an apparatus; a system; and/or a composition of matter; a computer program product embodied on a computer readable storage medium; and/or a processor, such as a processor configured to execute instructions stored on and/or provided by a memory coupled to the processor. In this specification, these implementations, or any other form that the invention may take, may be referred to as techniques. In general, the order of the steps of disclosed processes may be altered within the scope of the invention. Unless stated otherwise, a component such as a processor or a memory described as being configured to perform a task may be implemented as a general component that is temporarily configured to perform the task at a given time or a specific component that is manufactured to perform the task. As used herein, the term 'processor' refers to one or more devices, circuits, and/or processing cores configured to process data, such as computer program instructions.

[0022] A detailed description of one or more embodiments of the invention is provided below along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications and equivalents. Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

[0023] Embodiments described herein allow for a screw to easily be removed if it is stripped, broken, or otherwise damaged. This can save the patient's precious biomaterial by avoiding the need to remove the material surrounding the damaged screw in order to free it, and/or prevent the need to utilize extractor kits, or extensive procedures. Additionally, embodiments described herein allow more precise insertion techniques while reducing the risk of dropping or fumbling pieces of hardware in the operating room during a surgical procedure. Alongside external threads, the screw includes an internally threaded chamber that allows the screw to be removed without relying only on the screw head. In some embodiments, the external threads are right-handed, while the internal threads are left handed. The right-handed external threads and left-handed internal threads hold a dual purpose for both removal and insertion.

[0024] In some embodiments, a fastener device (e.g., a screw) includes a proximal head, a shaft extending from the proximal head and having external threads on at least a portion of the shaft, and a chamber extending internally from the proximal head into the shaft, at least a portion of the chamber having internal threads. The external threads have a first handedness and the internal threads have a second handedness different from the first handedness. For example, by having different handedness, it allows a tool engaged with the internal threads to remain engaged with the fastener when being turned in the direction to tighten the tool within the fastener's internal chamber while the external threads of the fastener being turned in the same direction are being loosened (backed out) from a patient during orthopedic and other surgical procedures. Additionally, when screwing in the fastener to a patient, the internal threads can be used to retain the fastener on to the tool in a more secure manner as compared to traditional press fit or tacking techniques. Other embodiments do not need to achieve all these objectives, and the claims hereof should not be limited to structures or methods capable of achieving these objectives.

[0025] In an example screw insertion, an interlock driver tool (e.g., driver bit) is secured into the left-handed internal threads of the screw by rotating counterclockwise. Once the driver tool is fully seated and preloaded to a set number, the provider will rotate the screw clockwise, interfacing the external right-handed threads with the bone. The opposing torque created by the external threads when being seated into the bone will unscrew the interlock driver tool from the internal left-handed threads and the tool will be removed from the screw. Once the interlock driver tool is removed from the screw, a secondary driver tool is utilized to interface with the head of the screw and rotated in the clockwise direction to complete the insertion and fasten the screw to a desired tightness. In an example screw removal, an interlock driver tool (e.g., driver bit) is secured into the left-handed internal threads of the screw by rotating counterclockwise. Once the driver tool is fully seated, the continued counterclockwise rotation will unscrew the screw's external right-handed threads and allow for the complete removal of the screw.

[0026] In some embodiments, features of the opposing handedness thread designs of the internal threads and the external threads are governed by a relative torque relationship. The torque relationship provides a calculated approach to how the internal and external threads relate to each other when subjected to insertion and removal torque values. To achieve desired torque values at a given load, the thread pitch and thread profile dimensions are selected so that the mechanical advantage and frictional conditions produce the desired torque relationship.

FIGS. 1A-1C are diagrams illustrating various views of an embodiment of a fastener. FIGS. 1D-1E are diagrams illustrating section views of an embodiment of the fastener. FIG. 1F is a diagram illustrating example dimensions of an embodiment of the fastener. Fastener **100** (e.g., screw) includes head **102**, shank **104**, external threads **106**, tip **108**, drive recess **110**, internal threads **112**, and shaft **114**. In some embodiments, fastener **100** is made of titanium, tin, brass, bronze, copper, aluminum, nickel, zinc, iridium, or another non-ferrous metal. In some embodiments, fastener **100** is made of a biocompatible polymer or plastic. In some embodiments, fastener **100** is made of stainless steel, steel, or another ferrous metal. In various embodiments, head **102** may be one of the following types: flat, round, locking, headless, low profile, or pedicle. In an alternative embodiment, head **102** is shaped as an external drive, allowing a socket tool or an external gripping tool to engage head **102** to turn fastener **100**. Examples of external drive head forms include hex, square, pentagon, star/torx or 12 point. Shank **104** is a smooth unthreaded portion between head **102** and portion of external threads **106** that may improve shear strength and prevent over-tightening of fastener **100**. Shank **104** is optional in various other embodiments (e.g., entire body of fastener is threaded). In some embodiments, handedness of external threads **106** is right-handed (e.g., threads spiral in a clockwise direction so that the fastener is tightened by turning the fastener clockwise). In some embodiments, handedness of external threads **106** is left-handed (e.g., threads spiral in a counterclockwise direction so that the fastener is tightened by turning the fastener counterclockwise). Although shape/threads of a cancellous screw are shown in external threads **106**, other types of threads may be utilized in other embodiments (e.g., cortical, lag, compression, fenestrated, etc.). Tip **108** may be self-tapping, self-drilling, or non-self-tapping in various embodiments. In some embodiments, shaft **114** includes a hollow center tunnel that spans the entire length of the fastener (e.g., cannulated screws).

[0027] Drive recess **110** includes an internal recess on a top of fastener **100** shaped/configured to engage a drive tool to rotate fastener **100**. Although an internal hex drive type is shown in drive recess **110**, drive types/shapes in other embodiments include slotted, cruciform, square, pentalobular, hexalobular (e.g., star/torx), triangular, or tamper-resistant. Drive recess **110** forms an opening part of a cavity with internal threads **112**. Handedness of internal threads **112** on walls of the cylindrical cavity is opposite the handedness of external threads **106**. In some embodiments, when handedness of external threads **106** is right-handed, the handedness of internal threads **112** is left-handed (e.g., threads spiral in a counterclockwise direction so that a tool engaged with the internal threads is loosened by turning the tool clockwise and tightened by turning the tool

counterclockwise). In some embodiments, when handedness of external threads **106** is left-handed, the handedness of internal threads **112** is right-handed (e.g., threads spiral in a clockwise direction so that a tool engaged with the internal threads is tightened by turning the tool counterclockwise and tightened by turning the tool clockwise). The chamber with internal threads **112** is closed on the bottom and open on the top through drive recess **110**. Because biomaterial can accumulate inside this partially enclosed cavity, after installation of the fastener in a subject, the cavity with internal threads **112** may be in part filled with hydrogel or another soluble and/or biocompatible material to reduce the accumulation of biomaterial inside the cavity. During removal of the fastener, this filler material may be scraped out or removed prior to screwing in a driver tool into the cavity with internal threads **112**. In the absence of a filler material, a pick tool may be utilized during the removal of the fastener to scrape out accumulated biomaterial. In an alternative embodiment, the cavity with internal threads is also open on the bottom so that a hollow opening spans the entire length of the fastener from the head to the tip of the fastener (e.g., cannulated screw). Dimensions (e.g., millimeter dimensions and/or angles) shown in FIG. 1F are merely examples and fasteners with different dimensions exist in other embodiments. In various embodiments, the diameter of the fastener (e.g., distance between external threads) is greater or equal to 6 mm.

[0028] FIG. 2A is a diagram illustrating an embodiment of an interlock driver tool. FIG. 2B is a diagram illustrating example dimensions of the interlock driver tool. Interlock driver tool **200** (e.g., driver bit) includes distal tip **202** with threads, neck portion **204**, and shank **206**. Threads of tip **202** are configured to cooperate with internal threads **112** of fastener **100** shown in FIGS. 1A-1F. Neck portion **204** is tapered and shaped to not engage drive recess **110** of fastener **100** when tip **202** is engaged with internal threads **112**. Shank **206** is configured to engage a chuck or a driver bit retainer of a driver power or hand tool. The shape/configuration of shank **206** provides a secure connection between interlock driver tool **200** and the power/hand tool (e.g., screwdriver) and transmits the rotational force from the power/hand tool to the working tip **202**. Although shown shank **206** is of a hex type, other embodiments may be of a different shank type (e.g., SDS, straight, etc.) or designed as a manual screwdriver. Dimensions (e.g., millimeter dimensions and/or angles) shown in FIG. 2B are merely examples and tools with different dimensions exist in other embodiments.

[0029] FIGS. 3A-3C are diagrams illustrating examples of engaging an interlock driver tool with a fastener. Interlock driver tool **200** of FIGS. 2A-2B is screwed into fastener **100** of FIGS. 1A-1F. Tip **202** of interlock driver tool **200** is passed through drive recess **110** and engaged with internal threads **112** of the cavity of interlock driver tool **200** by screwing tip **202** into fastener **100** until tip **202** is fully seated in the cavity having internal threads **112**.

[0030] FIG. 4 is a diagram illustrating an embodiment of a head driver tool. Head driver tool **400** (e.g., driver bit) includes distal tip **402** and driver engagement shank **406**. Hex tip **402** is configured to cooperate with internal walls of drive recess **110** of fastener **100** shown in FIGS. 1A-1F.

Although tip **402** is of a hex shape/type, in other embodiments, the tip shape/type of the head driver tool in other embodiments includes slotted, cruciform, square, pentalobular, hexalobular (e.g., star/torx), triangular, or tamper-resistant. Shank **406** is configured to engage a chuck or a driver bit retainer of a driver power or hand tool. Shape/configuration of shank **406** provides a secure connection between head driver tool **400** and the power/hand tool (e.g., screwdriver) and transmits the rotational force from the power/hand tool to the working tip **402**. Although as shown shank **406** is of a hex type, other embodiments may be of a different shank type (e.g., SDS, straight, etc.) or designed as a manual screwdriver.

[0031] FIGS. 5A-5C are diagrams illustrating examples of engaging a head driver tool with a fastener. Head driver tool **400** of FIG. 4 is engaged with fastener **100** of FIGS. 1A-1F. Tip **402** of head driver tool **400** engages walls of drive recess **110** via press fit. Rotation of head driver tool **400** rotates fastener **100** to screw or unscrew fastener **100**.

[0032] FIGS. 6A-6C are diagrams illustrating various views of alternate embodiments of the fastener shown in FIGS. 1A-1F. FIG. 6B is a diagram illustrating a section view of the embodiment of the fastener. Fastener **600** (e.g., screw) includes head **602**, shank **604**, threads **606**, tip **608**, drive recess **610**, and internal threads **612**. As compared to fastener **100**, fastener **600** has a shorter shank portion, different thread type (i.e., cortical) and a different drive recess type (e.g., hexalobular, star/torx).

[0033] FIG. 7 is a diagram illustrating an embodiment of a head driver tool. Head driver tool **700** (e.g., driver bit) includes distal Hexalobular tip **702**, elongated neck portion **704**, and driver engagement shank **706**. Tool **700** is an alternate embodiment of head driver tool **400** of FIG. 4. Hexalobular tip **702** is configured to cooperate with internal walls of drive recess **610** of fastener **600** shown in FIGS. 6A-6C. Shank **706** is configured to engage a chuck or a driver bit retainer of a driver power or hand tool.

[0034] FIG. 8 is a diagram illustrating an embodiment of an interlock driver tool. Interlock driver tool **800** (e.g., driver bit) includes distal tip **802** with threads, elongated neck portion **804**, and driver engagement shank **806**. Threads of tip **802** are configured to cooperate with internal threads **112** of fastener **100** shown in FIGS. 1A-1F and/or internal threads **612** of fastener **600** shown in FIGS. 6A-6C. Tool **800** is an alternate embodiment of interlock driver tool **200** of FIG. 2.

[0035] FIG. 9 is a diagram illustrating an embodiment of a head driver tool. Head driver tool **900** (e.g., driver bit) includes distal hex tip **902**, elongated neck portion **904**, and driver engagement shank **906**. Tool **900** is an alternate embodiment of head driver tool **400** of FIG. 4 with elongated neck portion **904**. Tip **902** is configured to cooperate with internal walls of drive recess **110** of fastener **100** shown in FIGS. 1A-1F. Shank **906** is configured to engage a chuck or a driver bit retainer of a driver power or hand tool.

[0036] FIGS. 10A-10P are diagrams illustrating various views of different embodiments of fasteners with secondary drive features. The various different embodiments of shown external and recessed drive features are merely examples, and other shapes, locations, configurations, and instance numbers of drive features exist in other embodiments.

[0037] FIGS. 10A-10B are diagrams illustrating various views of an embodiment of fastener **1010** with half-moon external drive features. FIGS. 10C-10D are diagrams illustrating section views of the embodiment of fastener **1010**. In some embodiments, fastener **1010** includes the features of fastener **100** shown in FIGS. 1A-1F. In addition to these features shared with fastener **100**, fastener **1010** includes half-moon external drive features **1012** shaped on the sides of the fastener's proximal head. Although two instances of external drive features **1012** are shown, any number of these external drive features may exist in other embodiments. In some embodiments, half-moon external drive features **1012** are utilized to further secure the fastener to a tool (e.g., screwdriver, power tool, etc.) utilized to rotate/drive the fastener. For example, features **1012** function as external head drive features to engage corresponding surface(s) and/or protrusion(s) of a tool to secure fastener **1010** to the tool when fastener **1010** is being rotated via the tool. In some embodiments, external drive features **1012** are utilized in conjunction with center drive recess **1014** and/or internal threads **1016** when securing fastener **1010** to the tool utilized to rotate/drive the fastener. For example, the tool includes a first mechanism to engage center drive recess **1014** and/or internal threads **1016** in conjunction with a second mechanism to engage external drive features **1012**.

[0038] FIGS. 10E-10F are diagrams illustrating various views of an embodiment of fastener **1020** with flat external drive features. FIGS. 10G-10H are diagrams illustrating section views of the embodiment of fastener **1020**. In some embodiments, fastener **1020** includes the features of fastener **100** shown in FIGS. 1A-1F. In addition to these features shared with fastener **100**, fastener **1020** includes flat external drive features **1022** shaped on the sides of the fastener's proximal head. Although two instances of external drive features **1022** are shown, any number of these external drive features may exist in other embodiments. In some embodiments, flat external drive features **1022** are utilized to further secure the fastener to a tool (e.g., screwdriver, power tool, etc.) utilized

to rotate/drive the fastener. For example, features **1022** function as external head drive features to engage corresponding surface(s) and/or protrusion(s) of a tool to secure fastener **1020** to the tool when fastener **1020** is being rotated via the tool. In some embodiments, external drive features **1022** are utilized in conjunction with center drive recess **1024** and/or internal threads **1026** when securing fastener **1020** to the tool utilized to rotate/drive the fastener. For example, the tool includes a first mechanism to engage center drive recess **1024** and/or internal threads **1026** in conjunction with a second mechanism to engage external drive features **1022**.

[0039] FIGS. **10I-10J** are diagrams illustrating various views of an embodiment of fastener **1030** with triangular external drive features. FIGS. **10K-10L** are diagrams illustrating section views of the embodiment of fastener **1030**. In some embodiments, fastener **1030** includes the features of fastener **100** shown in FIGS. **1A-1F**. In addition to these features shared with fastener **100**, fastener **1030** includes triangular external drive features **1032** shaped on the sides of the fastener's proximal head. Although two instances of external drive features **1032** are shown, any number of these external drive features may exist in other embodiments. In some embodiments, triangular external drive features **1032** are utilized to further secure the fastener to a tool (e.g., screwdriver, power tool, etc.) utilized to rotate/drive the fastener. For example, features **1032** function as external head drive features to engage corresponding surface(s) and/or protrusion(s) of a tool to secure fastener **1030** to the tool when fastener **1030** is being rotated via the tool. In some embodiments, external drive features **1032** are utilized in conjunction with center drive recess **1034** and/or internal threads **1036** when securing fastener **1030** to the tool utilized to rotate/drive the fastener. For example, the tool includes a first mechanism to engage center drive recess **1034** and/or internal threads **1036** in conjunction with a second mechanism to engage external drive features **1032**.

[0040] FIGS. **10M-10N** are diagrams illustrating various views of an embodiment of fastener **1040** with recessed pin drive features. FIGS. **10O-10P** are diagrams illustrating section views of the embodiment of fastener **1040**. In some embodiments, fastener **1040** includes the features of fastener **100** shown in FIGS. **1A-1F**. In addition to these features shared with fastener **100**, fastener **1040** includes recessed arc-shaped pin drive features **1042** recessed on the top of the fastener's proximal head offset from center drive recess **1044**. Although two instances of recessed arc-shaped pin drive features **1042** are shown, any number of these recessed drive features may exist in other embodiments. In some embodiments, recessed arc-shaped pin drive features **1042** are utilized to further secure the fastener to a tool (e.g., screwdriver, power tool, etc.) utilized to rotate/drive the fastener. For example, features **1042** function as additional recessed head drive features to engage corresponding pins(s) and/or protrusion(s) of a tool to secure fastener **1040** to the tool when fastener **1040** is being rotated via the tool. In some embodiments, recessed arc-shaped pin drive features **1042** are utilized in conjunction with center drive recess **1044** and/or internal threads **1046** when securing fastener **1040** to the tool utilized to rotate/drive the fastener. For example, the tool includes a first mechanism to engage center drive recess **1044** and/or internal threads **1046** in conjunction with a second mechanism to engage recessed arc-shaped pin drive features **1042**.

[0041] FIG. **11A** is a flowchart illustrating an embodiment of a process for inserting a fastener into a subject. FIG. **11B** is a diagram illustrating an example implementation of at least a portion of the process of FIG. **11A**. In some embodiments, at least a portion of the process of FIG. **11A** is performed during a surgical operation associated with one or more of the following: lumbosacral, acetabulum, hip, pelvis, thoracic, or spinal. In some embodiments, the process of FIG. **11A** is used to insert fastener **100** of FIGS. **1A-1F**, fastener **600** of FIGS. **6A-6C**, fastener **1010** of FIGS. **10A-10D**, fastener **1020** of FIGS. **10E-10H**, fastener **1030** of FIGS. **10I-10L**, and/or fastener **1040** of FIGS. **10M-10P** to a subject.

[0042] At **1102**, a first tool is coupled or mated onto internal threads of a fastener. This coupling or mating may be achieved by rotating the first tool and/or the fastener such that with respect to the fastener, the first tool is rotated in a first direction to screw a tip of the first tool into the fastener. Examples of the first tool include interlock driver tool **200** shown in FIGS. **2A-2B** or interlock

driver tool **800** shown in FIG. **8**. In some embodiments, the first tool (e.g., interlock driver bit) is coupled or mated to a surgical electronic/manual driver operated by a surgeon to rotate the first tool. An example of step **1102** is shown in setup step **1** of FIG. **11B** where the interlock driver is inserted into the screw's left-handed internally threaded chamber by rotating it counterclockwise. [0043] At **1104**, the first tool is rotated in a second direction to cause the fastener to rotate in the second direction to at least partially screw the fastener into a subject. The second direction is opposite the first direction. At **1106**, the first tool is removed from the fastener by continuing to rotate the first tool in the second direction and leaving the fastener at least partially inserted in the subject. An example of steps **1104** and **1106** are shown in insertion step **2** of FIG. **11B** where the screw coupled or mated to the interlock driver tool is placed at a desired position on the subject, and the screw is inserted by rotating the screw via the interlock driver tool clockwise into the subject (e.g., bone). Once a few of the external threads of the screw have been seated into the subject, the opposing torques from the turning of the interlock driver tool and the resistance of being inserted in the subject unscrews the interlock driver tool from the internal threads of the screw.

[0044] At **1108**, a second tool is interfaced with the fastener to drive the fastener in the second direction to further screw the fastener into the subject. Examples of the second tool include fastener head driver tool **400** shown in FIG. **4**, fastener head driver tool **700** shown in FIG. **7**, or fastener head driver tool **900** shown in FIG. **9**. In some embodiments, the second tool is interfaced with the fastener via a drive recess or an external or recessed drive configuration of the fastener. In some embodiments, the second tool (e.g., fastener head driver bit) is coupled or mated to a surgical electronic/manual driver operated by a surgeon to rotate the second tool. An example of step **1108** is shown in insertion tighten step **3** of FIG. **11B** where once the interlock driver tool is removed from the screw, a secondary driver is utilized to interface with the head of the screw and rotated in the clockwise direction to complete the insertion and fasten the screw to a desired tightness. In some embodiments, because biomaterial can accumulate inside the internally threaded cavity of the fastener, the cavity with the internal threads may be in part filled with hydrogel or another soluble and/or biocompatible material to reduce the accumulation of biomaterial inside the cavity.

[0045] FIG. **12A** is a flowchart illustrating an embodiment of a process for removing a fastener from a subject. FIG. **12B** is a diagram illustrating an example implementation of at least a portion of the process of FIG. **12A**. In some embodiments, at least a portion of the process of FIG. **12A** is performed during a surgical operation associated with one or more of the following: lumbosacral, acetabulum, hip, pelvis, thoracic, or spinal. In some embodiments, the process of FIG. **12** is used to remove fastener **100** of FIGS. **1A-1F**, fastener **600** of FIGS. **6A-6C**, fastener **1010** of FIGS. **10A-10D**, fastener **1020** of FIGS. **10E-10H**, fastener **1030** of FIGS. **10I-10L**, and/or fastener **1040** of FIGS. **10M-10P** from a subject.

[0046] At **1202**, a first tool is coupled or mated onto internal threads of a fastener inserted in a subject. This coupling or mating may be achieved by rotating the first tool such that with respect to the fastener, the first tool is rotated in a first direction to screw a tip of the first tool into the fastener. Internal threads of the fastener can be utilized for fastener removal in cases where a stripped, broken (e.g., a screw whose head is broken off), or otherwise damaged surgical screw needs to be removed from the body. The internal threads allow screws to be removed without having to drill into the patient's biological material that surrounds the screw and/or prevents the need to utilize extractor kits, or other extensive procedures. With respect to the fastener, the first tool is rotated in a first direction to screw a tip of the first tool into the fastener. Examples of the first tool include interlock driver tool **200** shown in FIGS. **2A-2B** or interlock driver tool **800** shown in FIG. **8**. In some embodiments, the first tool (e.g., interlock driver bit) is coupled or mated to a surgical electronic/manual driver operated by a surgeon to rotate the first tool. In some embodiments, prior to coupling or mating the first tool onto the internal threads of the fastener, accumulated biomaterial or prior placed filler material (e.g., hydrogel or another soluble and/or

biocompatible material) is removed (e.g., using a pick tool) from a cavity having the internal threads of the fastener. An example of step **1202** is shown in setup step **1** of FIG. **12B** where the interlock driver is inserted into the screw's left-handed internally threaded cavity by rotating it counterclockwise.

[0047] At **1204**, the rotation of the first tool is continued in the first direction to unscrew the external threads of the fastener from the subject. Given the opposing handedness of the internal threads and the external threads of the fastener, the direction of rotation of the first tool to insert it into the fastener is the same rotational direction to unscrew the fastener from the subject. With continual rotation, the first tool becomes fully engaged and bottoms out inside the fastener cavity with the internal threads, and the rotation of the first tool causes the fastener to rotate and become unscrewed from the subject. An example of step **1204** is shown in removal step **2** of FIG. **12B** where the continued rotation of the interlock driver in the counterclockwise direction causes the screw to become loosened until it is removed from the subject. Step **3** of FIG. **12B** shows that after removal, the screw can be safely disposed. The use of the first tool and the internal threads of the fastener to remove the fastener is especially useful when a drive recess of the fastener is damaged, stripped, or worn, preventing full engagement of a second tool configured to engage the drive recess of the fastener.

[0048] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There are many alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

Claims

1. A fastener device, comprising: a proximal head; a shaft extending from the proximal head and having external threads on at least a portion of the shaft; and a chamber extending internally from the proximal head into the shaft, at least a portion of the chamber having internal threads; wherein the external threads have a first handedness, and the internal threads have a second handedness different from the first handedness.
2. The device of claim 1, wherein the first handedness is right-handed and the second handedness is left-handed.
3. The device of claim 1, wherein the fastener device is coupled or mated to an interlock driver tool including distal external threads cooperating with the internal threads of the chamber.
4. The device of claim 3, wherein the fastener device and the interlock driver tool are configured to unscrew from each other while the fastener device is being screwed at least partially into a subject.
5. The device of claim 3, wherein the fastener device and the interlock driver tool are configured to be seated together via the internal threads while the fastener device is being rotated for removal from a subject.
6. The device of claim 1, wherein the fastener device is coupled or mated to a tool cooperating with the proximal head.
7. The device of claim 6, wherein the proximal head includes a drive recess and the tool cooperates with the proximal head via the drive recess.
8. The device of claim 1, wherein the proximal head includes a drive recess that is one or more of the following shapes/types: slotted, cruciform, square, pentalobular, hexalobular, triangular, or tamper-resistant.
9. The device of claim 1, wherein the fastener device is made of non-ferrous material.
10. The device of claim 1, wherein the fastener device includes a self-tapping tip or a self-drilling tip.
11. The device of claim 1, wherein the external threads are of a cortical or cancellous thread type.
12. The device of claim 1, wherein the fastener device is a part of a pedicle screw.

- 13.** The device of claim 1, wherein the fastener device is a cannulated screw with a hollow core.
- 14.** The device of claim 1, wherein the chamber is at least partly filled or covered with a hydrogel or soluble material.
- 15.** The device of claim 1, wherein the proximal head includes an external drive feature or a plurality of recessed drive features.
- 16.** A method, comprising: coupling or mating a first tool onto internal threads of a fastener via a rotation associated with a first rotation direction of the first tool with respect to the fastener, wherein the fastener includes a head, the internal threads, and external threads, and the external threads have a first handedness, and the internal threads have a second handedness different from the first handedness; rotating the first tool in a second rotation direction to cause the fastener to rotate in the second rotation direction to at least partially screw the fastener into a subject; removing the first tool from the fastener by continuing to rotate the first tool in the second rotation direction and leaving the fastener at least partially inserted in the subject; and interfacing a second tool with the fastener and driving the fastener in the second rotation direction to further screw the fastener into the subject.
- 17.** The method of claim 16, wherein the first rotation direction is counterclockwise, and the second rotation direction is clockwise.
- 18.** The method of claim 16, wherein the subject is a bone associated with one or more of the following: lumbosacral, acetabulum, hip, pelvis, thoracic, or spinal.
- 19.** A method, comprising: coupling or mating a first tool onto internal threads of a fastener inserted in a subject via a rotation associated with a first rotation direction of the first tool with respect to the fastener, wherein the fastener includes a head, the internal threads, and external threads, and the external threads have a first handedness, and the internal threads have a second handedness different from the first handedness; and continuing to rotate the first tool in the first rotation direction to unscrew the external threads of the fastener from the subject.
- 20.** The method of claim 19, wherein a drive recess of the fastener is damaged, stripped, worn, or otherwise damaged, preventing the drive recess from fully engaging a second tool configured to engage the drive recess of the fastener.
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